Uniwersytet SWPS

Task list no. 10 – In this lab, we will implement the longest common subsequence problem using dynamic programming and the Activity Selection algorithm using the greedy method.

Task1. Implement the DP algorithm for the LCS (Longest Common Subsequences) problem, and print the final LCS length table (c), direction table (b), and the LCS of the following strings.

String 1: A T C G G C T A A T C G A T C G G C T A A T C G A T C G G

String 2: T C G G C T A A T C G A T C G G C T A A T C G A T C G G C

Task 2. Develop and implement a greedy algorithm to solve the Activity Selection problem. Run your program on the provided activity list to find the maximum set of mutually compatible activities. Print selected activities with the start and finish times.

Activity number	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12	A13	A14	A15
Start time	1	3	0	5	3	5	6	8	8	2	12	14	15	18	19
Finish time	4	5	6	7	9	9	10	11	12	14	16	17	19	21	23

Note. The lab report must contain the implemented code along with the corresponding results.

Note. ALL PROGRAMS SHOULD COMPILE WITHOUT ERROR AND WORK (i.e. DO EXACTLY WHAT IS DESCRIBED IN THE TASK)!!!

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